

Extreme Theory: Kolmogorov-Arnold Networks

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by Dr. Smolensky

Abstract

Inspired by the recent advent of Kolmogorov-Arnold Networks (KANs) as delineated by Liu et al. (2024), this paper endeavors to extend the foundational concepts of the Kolmogorov-Arnold (KA) representation theorem into an "extreme theory" framework for neural network architectures. This approach leverages a rigorous, deductive process from the axiomatic underpinnings of the KA theorem, aiming to refine and expand the theoretical and practical utility of KANs. Unlike conventional Multi-Layer Perceptrons (MLPs) that employ fixed activation functions, KANs are designed with learnable activation functions on the network edges, replacing linear weight matrices with parameterized univariate functions, thereby enhancing model interpretability, compactness, and accuracy. This paper applies extreme theory to recent insights from Liu et al. (2024) KANs by systematically building upon the KA representation theorem, which was once seen as practically implausible in such applications. As such, the works this paper applies will be purely based on the recent paper by Liu et al. (2024) and the base Kolmogorov-Arnold Representation Theorem (find citations at end). Finally, Liu et al. (2024) proposing applicable KANs was released ~1 week prior to this work therefore I would like to emphasize that this is a continuing/developing understanding of the KA theorem and the KAN, not just for myself but the field as a whole.

(1) Identify fundamental dimensions.

Kolmogorov-Arnold Representation Theorem:

- a. *Multivariate Decomposition:* The theorem states that any multivariate continuous function can be expressed as a finite superposition of continuous functions of a single variable and addition. Specifically, for a smooth $f : [0, 1]^n \rightarrow \mathbb{R}$;

$$f(\mathbf{x}) = f(x_1, \dots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$$

where Φ_q and $\phi_{q,p}$ are univariate continuous functions.

- b. *Boundedness and Continuity:* The functions involved are continuous, and the multivariate function itself is defined on a bounded domain, typically within a cube like $[0, 1]^n$.
- c. *Finite Superposition:* There is a specific structure to how these univariate functions are composed. The theorem asserts that you need at most $2n + 1$ auxiliary univariate functions to represent an n -dimensional function.

- d. *Addition as a Core Operation*: The theorem emphasizes that addition is the only "true" multivariate operation needed, with all other complexities expressible through univariate functions.

(2) Define basic derived variables.

- a. *Univariate functions* $\phi_{q,p}(x_p)$: These functions transform each input variable independently. They are derived from each input dimension and are the simplest non-trivial transformations of the input variables
- b. *Outer Functions* Φ_q : These functions combine the outputs of the univariate functions $\phi_{q,p}$ for each dimension p and each combination q . The total number of these functions is $2^n + 1$.
- c. *Summations* $\sum_{p=1}^n \phi_{q,p}(x_p)$: For each group of transformations q , the univariate functions are summed to form a new variable that captures a certain aspect of the multivariate interaction modeled by f .

(3) Identify central theoretical constructs/concepts C.

- a. *Representation of Multivariate Functions*: The core concept that a function of several variables can be expressed as a combination of simpler univariate functions linked by addition.
- b. *Superposition Principle*: This principle states that the complex behavior modeled by f can be decomposed into simpler, additive components. This is fundamental in reducing the complexity of multivariate functions to more manageable forms.
- c. *Finite Composition*: The assertion that this decomposition and representation can be achieved with a finite number of operations and transformations, specifically with $2^n + 1$ univariate functions and their additions, where n is the number of input variables. This introduces an upper bound on the complexity needed to represent any function within this framework.
- d. *Continuity and Smoothness*: While not an operation, the requirement that functions Φ and ϕ are continuous underpins the theorem's applicability and ensures that the function f behaves well under the transformations prescribed by the theorem.

(4) Identify broad-scope (theoretically-interpreted) empirical generalizations G concerning these concepts C.

- a. *Generalizability Across Domains*: The method of decomposing multivariate functions into univariate functions linked by addition can be applied universally across different domains, whether in physics, finance, biology, or data science. This universality suggests

that complex systems often have underlying simpler mechanisms that can be uncovered using this representation.

- b. **Additive Models in Statistics:** Empirically, additive models have been found effective in various statistical modeling tasks because they allow for the interpretation of the effect of each variable independently of others. This principle supports the effectiveness of models that assume features contribute linearly (or through simple transformations) to the output.
- c. **Model Complexity and Overfitting:** Empirical observations in model training (such as in neural networks) suggest that limiting the complexity of the model (in terms of the number of operations or transformations, akin to $2n + 1$ in the theorem) helps in avoiding overfitting. This aligns with the principle that a finite and often small number of transformations are sufficient to capture the behavior of complex functions.
- d. **Robustness in Applications:** In real-world applications, models that maintain continuity and smoothness are generally more robust to small perturbations in input data. This underpins many strategies in robust machine learning and signal processing where ensuring smooth responses to inputs is critical.

(5) Identify fundamental theoretical axioms A — ‘first principles’ — stated over the central concepts C.

a. **Representation of Multivariate Functions:**

- i. **Decomposability:** $\forall f: R^n \rightarrow R$, continuous on a compact set, $\exists \Phi_q, p, \Phi_q: R \rightarrow R$ such that;

$$f(\mathbf{x}) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$$

- ii. **Universality:** The representation is valid for all f in the set of continuous functions on $[0, 1]^n$.

b. **Superposition Principle:**

- i. **Linearity in Composition:**

$$f(\mathbf{x}) = \sum_{q=1}^{2n+1} (\text{linear combination of } \phi_{q,p}(x_p))$$

- ii. **Additive Sufficiency:** $f(\mathbf{x})$ is fully recoverable by additive operations of $\phi_{q,p}(x_p)$.

c. **Finite Composition:**

- i. **Finite Complexity:**

For $f: R^n \rightarrow R$, a maximum of $2n + 1$ functions Φ_q are required.

- ii. **Minimal Representation:**

$\min(\text{number of } \Phi_q) = 2n + 1$ for optimal decomposition.

d. ***Continuity and Smoothness:***

i. *Continuous Transformation:*

$\phi q, p, \Phi q$ are continuous functions.

ii. *Smooth Integration:*

The composition of $\phi q, p$ and Φq preserves continuity and smoothness of f .

(6) **Identify general derivative laws L.**

a. ***Law of Partial Derivatives:***

- i. Given the framework and the assumptions of univariate functions being differentiable, the rate at which the output changes in response to changes in each input is dependent on the derivative properties of the univariate functions used in the composition, and their collective interaction through the outer function.

b. ***Law of Composition and Superposition:***

- i. The stacking or layering of multiple sets of functions is possible within the framework of the theorem.
- ii. This enables the construction of more complex models or networks by repeating the application of the theorem's structure at each layer.
- iii. Each layer's output becomes the input for the next, much like layers in a neural network.

c. ***Law of Continuity and Boundedness:***

- i. If the univariate functions and their compositions are continuous over a bounded domain, the resulting multivariate function will also exhibit continuity and boundedness.
- ii. This is crucial for ensuring that the function behaves predictably and without abrupt changes over its input domain, which is important for both theoretical and practical applications.

d. ***Law of Uniform Convergence:***

- i. When sequences of univariate and outer functions converge uniformly to certain functions, the composite multivariate function constructed from these sequences will also converge uniformly.
- ii. This principle is particularly useful in numerical methods and simulations, where functions are approximated through discrete or finite representations.

(7) **Deduce laws L from axioms A.**

a. ***Partial Derivatives, with the axiomatic basis of Continuous Transformations and Finite Complexity:***

Given that each $\phi q, p$ and Φq are continuous (Axiom of Continuous Transformation), and assuming they are differentiable as a reasonable extension for practical applications, the partial derivative of f with respect to an input (xp) is given by the chain rule applied to the theorem's structure:

$$\frac{\partial f}{\partial x_p} = \sum_{q=1}^{2n+1} \Phi'_q \left(\sum_{p'=1}^n \phi_{q,p'}(x_{p'}) \right) \cdot \phi'_{q,p}(x_p)$$

Where the derivative of f w.r.t. one of its inputs x_p is the sum of the derivatives of the functions Φq evaluated at the inner sums, multiplied by the derivatives of the univariate functions $\phi q, p$ associated with x_p .

b. ***Composition and Superposition, with the axiomatic basis of Decomposability and Additive Sufficiency:***

Since f is decomposable into univariate functions linked by addition (Axiom of Decomposability), additional layers of functions can also be structured in the same manner, allowing for recursive or nested application of the theorem:

$$f(\mathbf{x}) = \sum_{q=1}^{2m+1} \Psi_q \left(\sum_{r=1}^m \theta_{q,r} \left(\sum_{p=1}^n \phi_{r,p}(x_p) \right) \right)$$

This law directly follows from the structure defined by the Axiom of Finite Complexity, where f is composed of $2n + 1$ univariate functions.

c. ***Law of Continuity and Boundedness, with the axiomatic basis of Continuous Transformation and Minimal Representation:***

If the transformations $\phi q, p$ and Φq are continuous over a compact domain, the composite function f inherits this continuity due to the properties of continuous functions under finite operations. This follows from general principles of continuity in calculus and the fact that a finite composition of continuous functions remains continuous.

d. ***Law of Uniform Convergence, with the axiomatic basis of Continuous Transformation and Decomposability:***

Uniform convergence of $\phi q, p(k)$ and $\Phi q(k) \Rightarrow$ uniform convergence of $f(k)$ to f . This leverages the finite, decomposable structure of f as per the Axiom of Decomposability, alongside the uniform convergence criteria.

(8) Formally deduce [as theorems] broad-scope empirical generalizations G from first principles A.

→ **Theorem 1:**

- ◆ *Universality of Function Representation* derived from Axioms of Universality and Decomposability. Every continuous multivariate function on a compact domain can be universally represented by a finite composition of univariate functions and additions.

→ **Theorem 2:**

- ◆ *Optimization of Model Complexity* derived from Axioms of Finite Complexity and Minimal Representation. The representation of any continuous multivariate function by exactly $2n + 1$ univariate functions optimizes computational efficiency and model complexity

→ **Theorem 3:**

- ◆ *Stability and Predictability in Applications* derived from Axioms of Continuous Transformation and Smooth Integration. Functions structured under the Kolmogorov-Arnold framework respond continuously to input variations, ensuring stable and predictable behavior in applications.

(9) Identify a class of basic systems Sp distinguished by their values of unobservable parameters p.

Class of Basic Systems:

- i. ***Number of Variables:*** The dimensionality of the input space can significantly affect the complexity and the number of required univariate functions.
- ii. ***Types of Univariate Functions:*** Deals with handling complex, high-dimensional data.
- iii. ***Smoothness and Continuity Requirements:*** Systems may vary based on the degree of smoothness required of the univariate functions, affecting their applicability in different scenarios.
- iv. ***Interactions Among Variables:*** Some systems may exhibit more complex interactions among variables, requiring a more intricate composition of the univariate functions.

(10) Derive predicted observable properties O_p of basic systems S_p as a function of their parameters p .

Parameters:

- i. **n :** Number of input variables
- ii. **$type$:** Types of univariate functions
- iii. **$smooth$:** Degree of smoothness or differentiability required in the univariate functions.
- iv. **$interact$:** Patterns of interactions among variables captured by the univariate functions.

Observable Properties:

The observable properties of a system can generally be expressed as:

$$O_p = f(n, type, smooth, interact)$$

where f is a function that maps the system parameters to the observable outcomes.

- i. ***Response Stability*:** Quantifies the system's complexity based on the number of input variables. (From the axiom of finite complexity, where the minimum number of univariate functions required is $2n + 1$.)
- ii. ***System Complexity*:** Related to the system's robustness. Coming from the axiom of continuous transformation, higher smoothness parameters imply more stable and predictable responses to input variations.
- iii. ***Interaction Sensitivity*:** The degree of non-linearity in the system's output, and is based on how the interactions are configured (e.g., linear vs. nonlinear interactions among variables), the sensitivity of the system to changes in specific variables can be observed and measured.

(11) Identify essential properties of a particular empirical system E_0 .

Given a generic empirical system E_0 , key properties include:

- i. ***System Dynamics*:** How the system evolves over time or in response to external inputs.
- ii. ***Input-Output Behavior*:** How inputs are transformed into outputs.
- iii. ***Complexity and Dimensionality*:** The number of variables and relationships that the system manages.
- iv. ***Robustness and Sensitivity*:** How sensitive the system is to variations in inputs or environmental conditions and its ability to maintain functionality under stress or disturbance.

- v. **Scalability:** How well the system can adapt or be scaled in terms of size, complexity, or scope.
- vi. **Noise and Error Tolerance:** The system's ability to handle inaccuracies or noise in data inputs without significant degradation in output quality.

(12) Identifying a Basic System $S\pi$ Idealizing $E0$

- a. **$E0$ represents a complex empirical system** in a field like data analytics, computational physics, or a dynamic control system where:
 - i. Input-output behavior is nonlinear and potentially involves high-dimensional data.
 - ii. System dynamics are critical, with a need for robust, time-sensitive responses.
 - iii. Stability and robustness are essential under varying operational conditions and input noise.
- b. **System Structure and Function:**
 - i. **Learnable Univariate Function Edges** that dynamically adapts, and is designed to transform individual input dimensions. These functions are parametrized typically as splines, providing a flexible mechanism to model complex nonlinear interactions inherent in $E0$.
 - ii. **Summation Nodes** perform summation of inputs from multiple edges, integrating transformed data without additional nonlinear processing at the node level.
 - iii. **No Linear Weights** focuses the learning and adaptability of the network on the edge functions $(\phi_{q,p})$, streamlining the network and potentially reducing computational overhead.
 - iv. **Flexibility in Depth and Width**, not limited to the original two-layer representation. This adaptability allows the network to be tailored to specific complexities and dimensions of input data, crucial for handling the high-dimensional datasets typical of $E0$

(13) Derive a formal model of $E0$ from the idealized basic system $S\pi$.

- a. **The formal model of $E0$ can be realized through a Kolmogorov-Arnold Network (KAN) model:**
 - i. **Edge Functions:** $\phi_{q,p} : R \rightarrow R$, parametrized by a set of coefficients determining the shape of the spline for each edge.
 - ii. **Aggregation at Nodes:** $\Phi_q(y_1, \dots, y_m) = \sum_{i=1}^m y_i$, where $y_i = \phi_{q,i}(x_i)$
 - iii. **Network Adaptability:** Network configurations can vary, with n input dimensions and m output dimensions, structured over potentially multiple layers.

- iv. **Absence of Linear Weights:** KANs do not use traditional weight matrices between layers, which typically involve linear transformations. Instead, all learning and adaptability are concentrated within the univariate functions on the edges, significantly reducing the computational overhead and focusing on capturing nonlinear dynamics effectively.
- b. *Given these components, the function f modeled by KAN for the empirical system $E0$ can be formally represented as:*

$$f(\mathbf{x}) = \sum_{q=1}^Q \Phi_q(\phi_{q,1}(x_1), \phi_{q,2}(x_2), \dots, \phi_{q,n}(x_n))$$

Where $x = (x_1, \dots, x_n)$ is the input vector, $\phi_{q,p}$ are the univariate functions applied at each edge, and Φ_q are the summation nodes that integrate these transformed inputs.

(14) Observe detailed properties D0 of the particular empirical system E0.

- a. **Interpretability:** One of the properties of KANs is their enhanced interpretability. KANs facilitate an intuitive understanding of the function mappings through their structure, where each edge function is represented as a learnable spline, making it easier to trace and understand the transformation applied at each step.
- b. **Accurate Low-Dimensional Approximations:** KANs excel in accurately representing functions, particularly in low-dimensional settings. This accuracy is achieved through the optimized use of spline functions, which are adept at capturing the nuances of complex nonlinear interactions with fewer parameters and less computational overhead.
- c. **Efficient Handling of High-Dimensional Data:** While KANs naturally address the curse of dimensionality better than MLPs by breaking down high-dimensional functions into compositions of one-dimensional functions, they also provide a scalable approach to managing high-dimensional inputs effectively, maintaining high accuracy with a relatively lower increase in parameter count.
- d. **Dynamic Adaptability:** KANs can dynamically adapt their structure—for instance, through the extension of spline grids—allowing the network to improve its approximation accuracy continuously without retraining from scratch.
- e. **Robustness to Overfitting:** By leveraging spline functions and maintaining a clear compositional structure, KANs demonstrate a robust performance profile, particularly in terms of generalization across different datasets and tasks. This robustness is further enhanced by their inherent interpretability, allowing for easier diagnostics and adjustments.

(15) Deduce the parameter values p of the basic system S_p that idealizes the particular system E_0 by fitting predictions Op to observations D_0 .

a. *Key Predictions Op and Observations D_0 :*

- i. **Prediction Op :** KANs' ability to provide accurate and interpretable models through the use of learnable spline-based univariate functions on edges and efficient summation operations at nodes.
- ii. **Observation D_0 :** E_0 's need for high interpretability, accuracy in low-dimensional approximations, efficient handling of high-dimensional data, dynamic adaptability, and robustness to overfitting.

b. *Parameter Values π :*

- i. **Type of Univariate Functions:** Choose spline functions as the univariate transformations on edges.
- ii. **Number of Spline Knots:** Adjust the number of knots in the spline functions to balance complexity and overfitting.
- iii. **Network Depth and Width:** Configure the number of layers (depth) and the number of nodes per layer (width).
- iv. **Learning Rate and Optimization Strategy:** Set the learning rate and choose an optimization strategy that supports the dynamic adaptation of spline coefficients during training.

(16) Use p to predict & test observable properties in Op beyond those of D_0 .

a. *Generalization Across Datasets:*

- i. **Prediction:** KANs will show superior generalization across various datasets, especially in complex pattern recognition or predictive modeling.
- ii. **Reported Result:** Liu et al. (2024) demonstrated enhanced generalization capabilities compared to traditional MLPs, particularly in scenarios involving non-standard data distributions and complex pattern recognition tasks.

b. *Computational Efficiency:*

- i. **Prediction:** KANs should be more computationally efficient due to their absence of weight matrices and streamlined function-based processing.
- ii. **Reported Result:** Liu et al. (2024) highlight that the absence of traditional weight matrices and the use of univariate spline functions reduce both the computational cost and memory usage during training and inference.

c. *Adaptability to Task Complexity:*

- i. **Prediction:** The adaptable network parameters allow KANs to effectively handle various levels of task complexity.

- ii. **Reported Result:** KANs were tested (Liu et al, 2024) on tasks with varying complexity levels, and the results showcased their ability to adapt effectively through adjustments in network configuration (depth and width), maintaining high performance across tasks.

d. Generalization Across Datasets:

- i. **Prediction:** KANs' reliance on univariate transformations should enhance their robustness to input noise and perturbations.
- ii. **Reported Result:** Liu et al. (2024) include tests that demonstrate the robustness of KANs against noise and other forms of input perturbations. The results confirm that KANs maintain stable and accurate outputs even under conditions of significant data variability.

Bibliography

Arnold, V. (1957). Representation of continuous functions of three variables by the superposition of continuous functions of two variables. *Springer EBooks, 1*, 47–133.

https://doi.org/10.1007/978-3-642-01742-1_6

Arnold, V. (1957). Representation of continuous functions of three variables by the superposition of continuous functions of two variables. *Springer EBooks, 1*, 47–133.

https://doi.org/10.1007/978-3-642-01742-1_6

Liu, Z., Wang, Y., Vaidya, S., Ruehle, F., Halverson, J., Soljačić, M., ... & Tegmark, M. (2024).

KAN: Kolmogorov-Arnold Networks. arXiv preprint arXiv:2404.19756.